

RapidSmith: An FPGA CAD Tool

RapidSmith is an XDL-based FPGA CAD tool written in Java for FPGAs from Xilinx. It is intended to serve as a rapid prototyping platform for research ideas and algorithms relating to low-level FPGA design

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Productivity of design engineers is significantly affected by the time-consuming process of field-programmable gate array (FPGA) compilation. Five engineers from the NSF Center for High-Performance Reconfigurable Computing (CHREC) presented a paper on rapid prototyping tools for FPGA designs, discussing the effect of slow FPGA compilation process on the designers' productivity. Their solution to this problem was a tool that they called RapidSmith.

RapidSmith is an open source software that lets you perform electronic design automation (EDA) tasks. It is free for both personal and commercial uses, thus the perfect choice for those who want an alternative for EDA programs.

What is RapidSmith

A research-based computer-aided design (CAD) tool for FPGAs, RapidSmith is based on the Xilinx Design Language (XDL). The software aims to be fully compatible with all the latest Xilinx FPGAs and works with Xilinx ISE 11.1 and higher versions. It

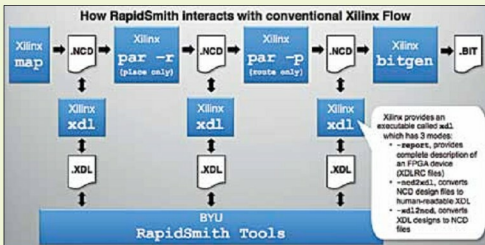


Fig. 1: How RapidSmith interacts with conventional Xilinx Flow

provides an easy-to-use rapid prototyping platform for new FPGA CAD algorithms and ideas, and it is also a useful infrastructure to demonstrate new and unique ideas on real Xilinx FPGAs.

What is XDL

XDL is Xilinx's gateway that provides RapidSmith to create tools and manipulate Xilinx designs at a very low level. It describes designs at the FPGA primitive level and can represent designs that are partially or fully unplaced and unrouted. Basically, XDL design

files have two common statements, 'instance' and 'net'.

Features

There are over 70 classes and 650 application programming interfaces (APIs) in RapidSmith, of which 225 APIs are in the design package alone. These APIs help manipulate designs in RapidSmith. Graphical user interfaces (GUI) are built with QT (qt jambi). It includes several javadocs and documentation inside the software as well as tools for comparing XDL designs. Moreover, there is excellent

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